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Faculty of Engineering
Tel Aviv University



FINAL PROJECT · No. 3221

Optical Fibers for Smart Cities

Fiber Optic (DAS) Sensing for Real Time Vehicle Tracking and Road Anomaly Detection

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School of Electrical Engineering, Tel Aviv University · 2025–2026

Project Topic

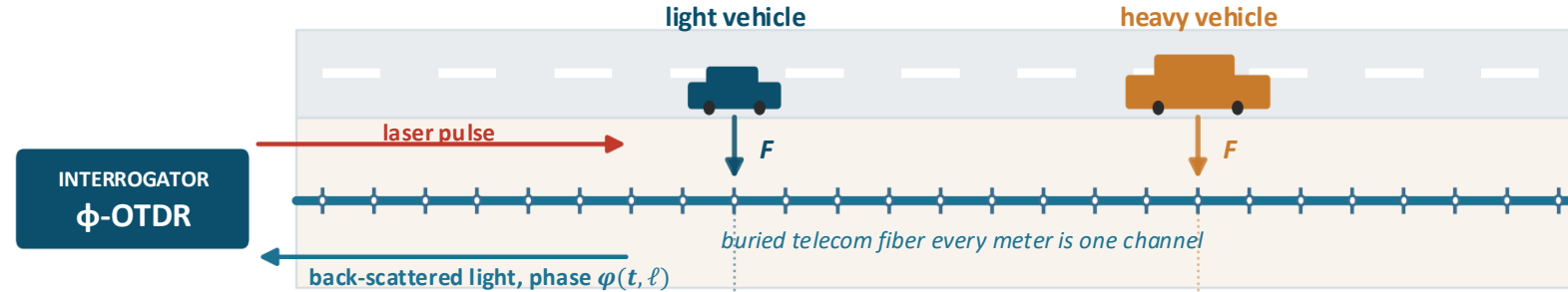
- Modern cities are laced with thousands of kilometers of telecommunication optical fiber, already deployed beneath most streets.
- Distributed Acoustic Sensing (DAS) transforms a standard fiber into a dense array of vibration sensors. each meter of fiber becomes an independent channel.
- A passing vehicle vibrates the ground and shifts the optical phase of the back-scattered light, giving a strain-rate signal whose amplitude encodes weight and slope encodes speed.
- **SimStudio is a complete desktop simulation, sensor-fusion and anomaly detection environment.**
 - Imports real OpenStreetMap networks, drives physically grounded vehicles, and synthesizes three modalities: DAS, camera, GPS.
 - Fuses them into stable vehicle tracks and detects anomalies from a read-only analytics layer.



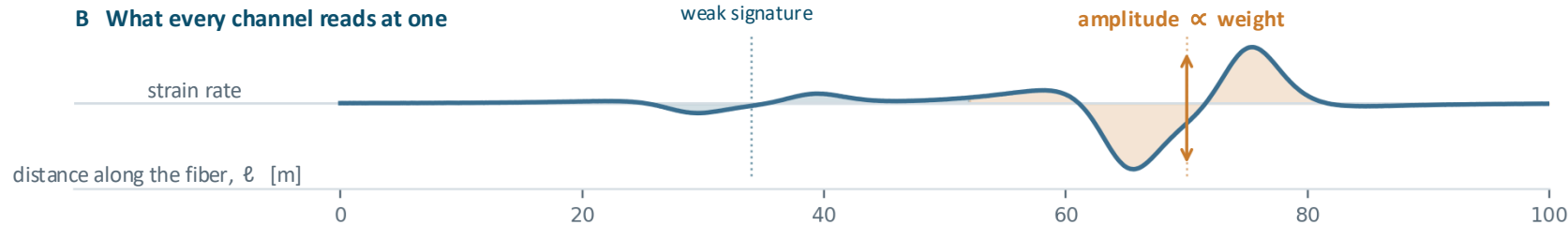
Real North Tel-Aviv road network — 144 simulation regions seed the normal-driving baseline.

How One Fiber Becomes Thousands of Sensors

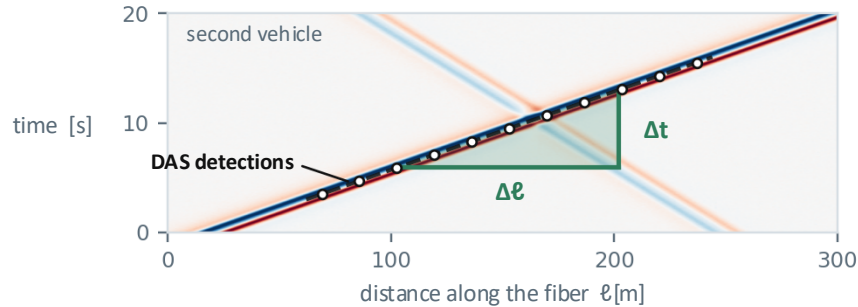
A One laser pulse in, back-scattered light out



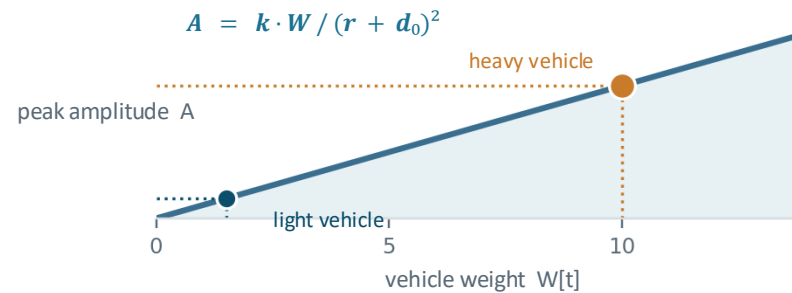
B What every channel reads at one



C Speed, the slope of the streak



D Weight, invert the amplitude decay



THE FORMAL MODEL

① Quasi-static ground displacement under a point load F (Flamant–Boussinesq):

$$u_x(x, y, z) = \frac{F}{4\pi G r^2} \left(\frac{x}{r} + \frac{2\nu - 1}{1 + \frac{z}{r}} \right)$$

② The fiber differences it over the gauge length L and reads the rate, what a channel records:

$$\dot{\epsilon}_{\text{DAS}} = \frac{1}{L} \left[\dot{u}_x \left(x + \frac{L}{2} \right) - \dot{u}_x \left(x - \frac{L}{2} \right) \right]$$

IN SIMSTUDIO WE INVERT THIS

SPEED, a vehicle lights up successive channels; the slope of the streak (panel C):

$$v = \frac{\Delta \ell}{\Delta t}$$

WEIGHT, invert the amplitude decay, corrected for the distance r to the fiber (panel D):

$$W = \frac{A(r + d_0)^2}{k}, \quad A = k \frac{W}{(r + d_0)^2}$$

- **position**, the channel index that fires
- **acceleration**, dv/dt along the track
- band < 1 Hz: the vehicle's load, not street noise

Engineering Motivation & Goals

- Re-use existing infrastructure. the fiber already exists, so no extra hardware is needed and incremental cost is low.
- Every meter of fiber is an independent channel. a passing vehicle leaves a moving signature whose amplitude encodes its weight and whose slope across channels encodes its speed.
- Continuous spatial coverage over tens of kilometers from a single cable, robust to lighting and weather.

Research question:

Can fusing DAS with camera and GPS reliably track individual vehicles across a road network, and detect anomalous behavior from the fused signal alone?

PROJECT TARGETS → ACHIEVED

Cross-street tracking

target $\geq 90\%$ accuracy

Sub-metre RMSE

Position / speed tolerance

target $\leq \pm 5$ m

Max error 1.52 m

Anomaly detection

target $\geq 5 / 10$ scenarios

6 of 9 detected

Kalman fusion success

target $\geq 90\%$

All 3 sensors fused

Simultaneous vehicles / segments

target $\leq 10 / \leq 5$

Run routinely

End-to-end latency

target ≤ 60 s

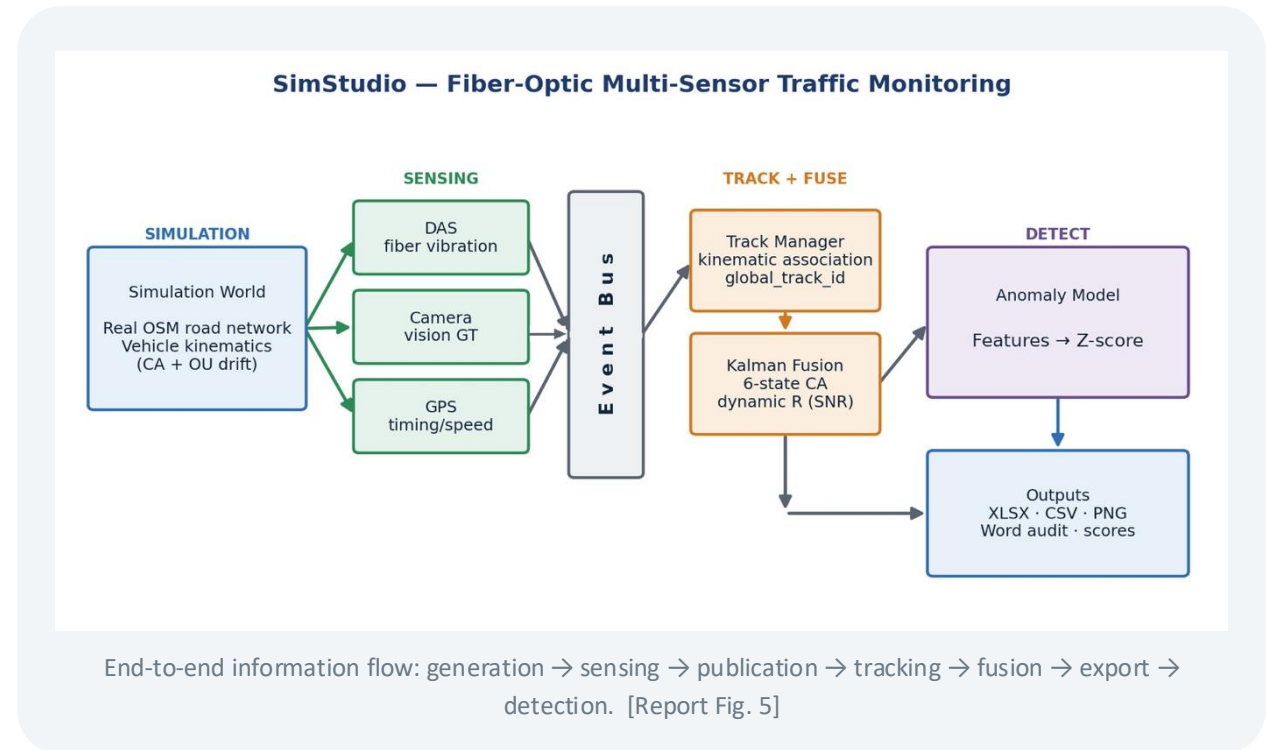
Profiling pending

Five targets met; only real-time latency (Partial) remains to be profiled.

System Architecture



- Implemented entirely in Python. every stage consumes only upstream outputs.
- The anomaly model never imports the simulator so it cannot benefit from information a real deployment would not have.
- Acyclic dependencies: every component is independently testable (pytest suite guards each boundary).
- 25 Hz simulation engine with a Tkinter desktop editor for interactive scene authoring.



DAS · primary

- Flamant–Boussinesq strain: $A = \frac{W}{(r+d_0)^2}$
- Detection sigmoid in SNR^2 ; each meter = 1 channel
- Along-fiber position & speed ($\sim 25\text{ Hz}$)

Camera

- Full 2-D (x, y) position, $\sim \text{frame rate}$
- Pinhole projection with stochastic FOV gating
- Noise: floor + range + off-axis angle

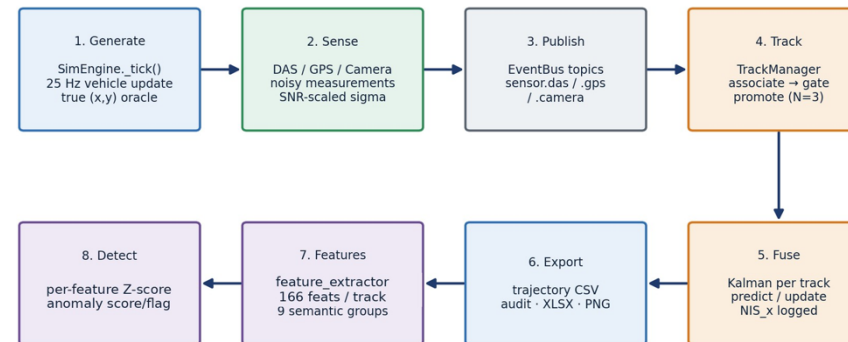
GPS

- Absolute position (x, y), $\sim 1\text{ Hz}$, $\sigma_0 \approx 2.5\text{ m}$
- Noise grows with distance from receiver
- Low rate and accuracy

SNR-DRIVEN MEASUREMENT NOISE

- Each sensor exposes a physically-derived uncertainty the Kalman filter consumes directly: $\sigma_{DAS} = \max\left(0.8, \frac{k_{das}}{\sqrt{SNR}}\right)$, scaled by confidence.
- So the filter trusts a heavy, nearby vehicle far more than a light, distant one automatically, with no manual tuning.
- DAS variance can be an order of magnitude smaller than GPS.

End-to-End Information Flow



TRACKING & KALMAN FUSION

- TrackManager keeps a stable global track ID across segment, kinematics only, never the oracle ID.
- Candidates scored by predicted-vs-measured properties. 3 agreeing hits to confirm so DAS ghosts rejected.
- 6-state constant-acceleration Kalman filter, decoupled x/y, Joseph-form covariance.
- a sustained rise above 1.0 is an anomaly.

Anomaly Detection Model

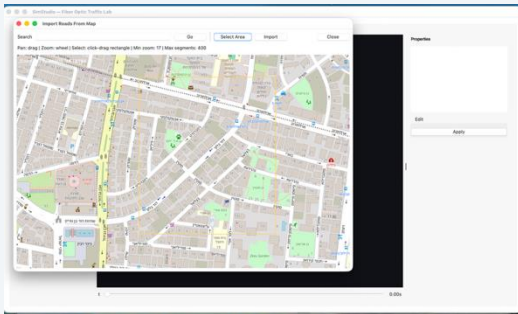
- **A read only, unsupervised model. it learns the distribution of normal driving and scores each vehicle by deviation.**
- Layer 1: Each vehicle trajectory → a 166-feature behavioral fingerprint across 9 semantic groups.
- **Layer 2: the per feature Z-score baseline flags a vehicle when its largest $|Z|$ exceeds the decision threshold, and reports which features drove the flag.**
- DAS weight is back-estimated from the Flamant–Boussinesq law to flag overweight or misclassified vehicles.
- CUSUM on NIS is a complementary temporal diagnostic: the Z-score says whether a vehicle is anomalous, CUSUM says when it started.

166 FEATURES PER VEHICLE · 9 GROUPS

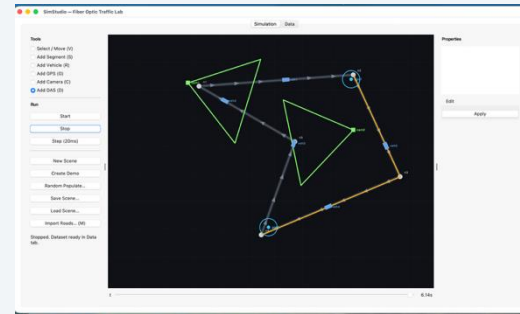
Coverage & sensor utility	54
Kinematics	35
Inter-vehicle interaction	20
Kalman-filter quality	13
Sensor disagreement	11
DAS signal quality	11
Camera & GPS	8
DAS weight estimation	3
Scenario context	1

156 features are scored by the model (identifiers & metadata excluded).

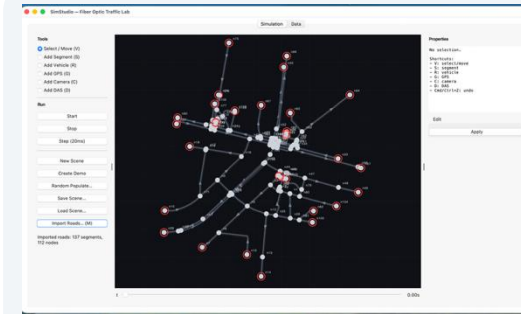
► LIVE DEMO



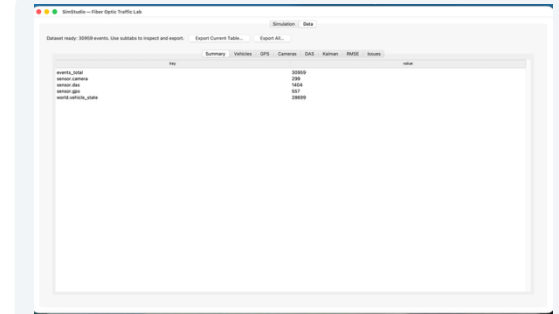
1 • Import real streets (OSM)



2 • Build & place sensors



3 • Simulate traffic (25 Hz)



4 • Inspect fused data streams

- Workflow: generate a scenario, run the simulation (optionally in batch), collect the exports, then run feature extraction and anomaly scoring.
- Each recorded run produces trajectories, a styled Excel workbook, an audit report and per-track visualizations, the same files are then fed to the anomaly layer later to score each vehicle.

Results Tracking Accuracy

0.35 m

mean track RMSE

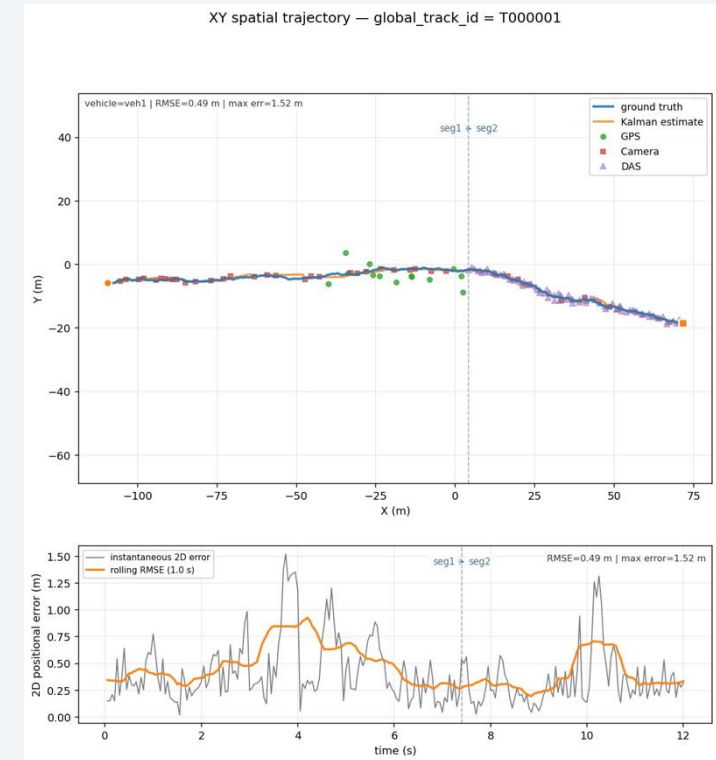
0.49 m

cross-segment RMSE

1.52 m

max error ($\leq \pm 5$ m)

- A vehicle fused across the seg1→seg2 boundary reached 0.49 m RMSE (max 1.52 m). the mean across the nine evaluation scenarios was 0.35 m (best 0.19, worst 0.45).
- DAS gives high rate, high trust updates along the fiber. the camera contributes accurate fixes at intersections. GPS supplies low-rate absolute corrections between DAS zones.
- A stable identity is maintained across every segment transition.
- Fixing a subtle GPS fragmentation or a ghost track bug with an identity aware ambiguity guard ensured GPS reached the correct confirmed track rather than orphan fragments, improving fusion accuracy.



Fused cross-segment track vs. ground truth.

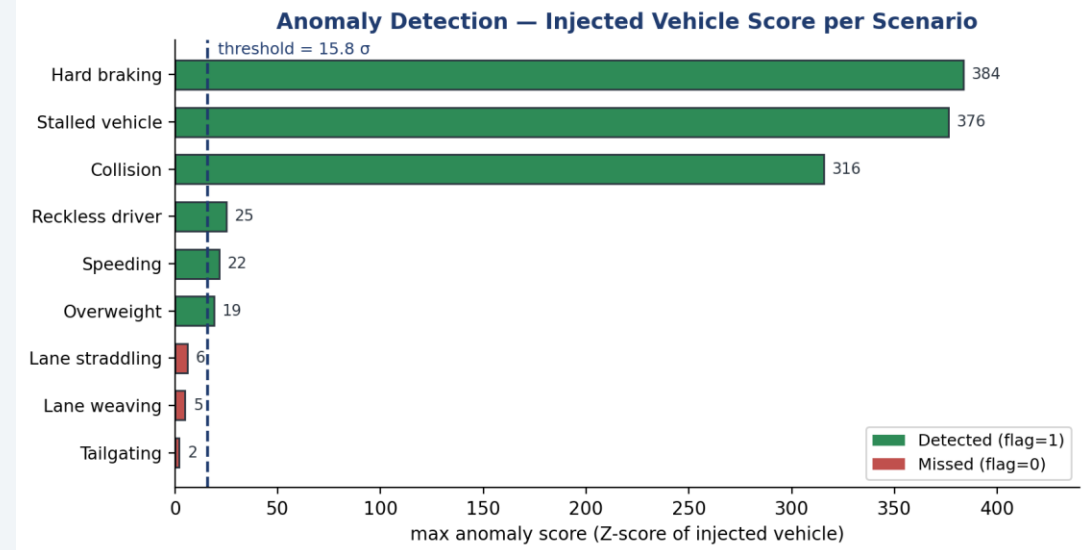
Results Anomaly Detection

6 / 9

anomalies detected

target $\geq 5 / 10$ — met

- Detected: hard braking, stalled vehicle, collision, reckless driving, speeding, overweight, longitudinal & sensor level anomalies separate clearly.
- Interpretable: for over speeding the flag was driven by relative speed at closest approach, mean speed, and lateral velocity.
- Missed: lane straddling (6.2 σ), weaving (5.0 σ), tailgating (2.1 σ), subtle behaviors below the conservative $\approx 15.4 \sigma$ threshold chosen to keep false positives low.



Injected-vehicle anomaly score per scenario vs. the conservative decision threshold (dashed).

- A real camera + DAS channel operated on an existing telecom fiber on Klausner St., beside the TAU campus.
- A city bus decelerating to a bus stop was replayed through the exact, unchanged pipeline on the true street geometry.
- Fused 84 DAS + 30 camera measurements over 41.8 s, ran on prediction alone for 42% of the run (longest gap 3.98 s) and never lost the track through the stop.

1.17 m

fused track RMSE

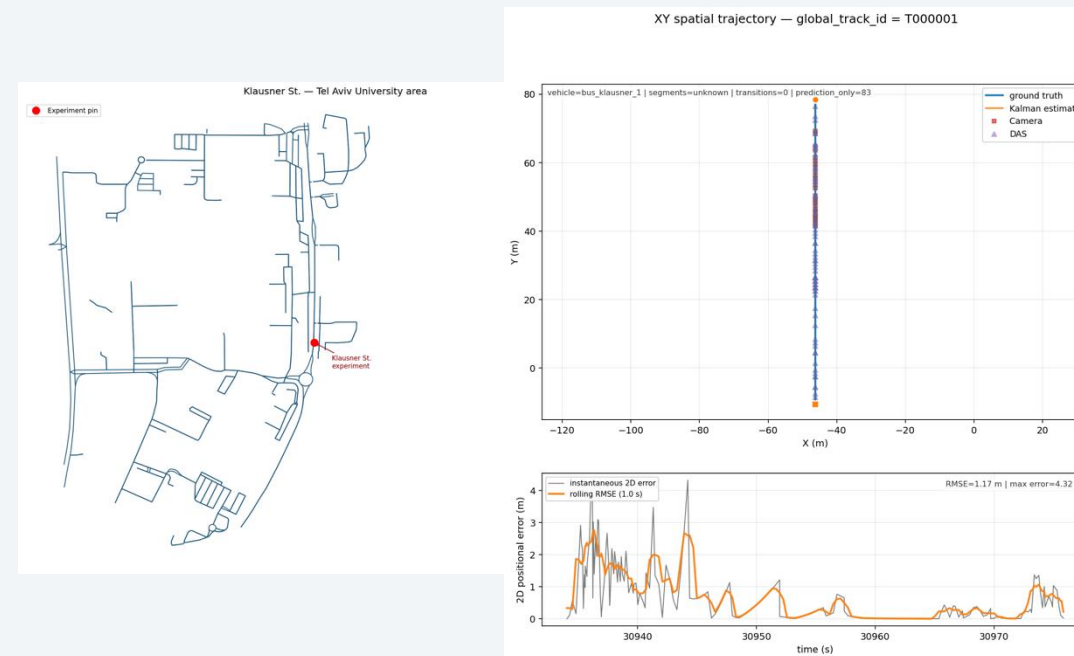
0.33 m

DAS per-sensor RMSE

2.49 m

camera per-sensor RMSE

The camera's measured error far exceeded its reported $\sigma = 0.2\text{m}$, exactly the failure mode the design anticipates. Cameras must be calibrated at each site before a larger real world deployment.



Klausner St. site (OSM) and the fused bus track through the stop.

SUGGESTIONS FOR CONTINUATION

1 Catch the tricky anomalies

Use the labelled examples we now have to train a smarter model, so it can spot the hard cases we miss today (weaving, straddling, tailgating).

2 Add better features

Measure side-to-side motion and how close cars get, and combine the 'what happened' and 'when it happened' signals so odd behaviour stands out.

3 Calibrate the cameras

Check each camera's real accuracy at its own spot before trusting it, so it helps the fusion instead of hurting it.

4 Test on real streets

Put a real DAS device on an instrumented street and run a longer real-world trial, building on our single Klausner recording.

5 Use the raw fiber signal

Work directly with the full DAS signal, not just simplified numbers, to detect events earlier and in more detail.

6 Run it live, city-wide

Make it fast enough for real time, follow cars across many intersections, and show live tracks and alerts on a dashboard.

PROJECT DOCUMENTATION

SimStudio

Engine & sensor models, event bus, track manager, Kalman fusion, export/audit, Tkinter editor; install script + packaged macOS app.

Anomaly model & tests

Feature extractor, CUSUM detector, per-feature Z-score scorer, batch scripts; pytest across simulator core, filter, tracker and export.

Data & repository

OSM networks, ~150 normal scenarios (12 conditions) + 9 injected-anomaly scenarios; README, changelog, validation protocol — github.com/noasapir3/simstudio-das-fusion

KEY REFERENCES

- Cohen, K., Hen, L., & Lellouch, A. "A fiber-optic traffic monitoring network trained with video inputs," Scientific Reports, vol. 15, art. 14928, 2025 (arXiv:2412.12743).
- Kalman, R. E. J. Basic Eng. 82(1):35–45, 1960. · Page, E. S. Biometrika 41(1–2):100–115, 1954. · Welch & Bishop, "An Introduction to the Kalman Filter," UNC. · OpenStreetMap contributors.